Dimension-Six Twisted Convolution as Arithmetic Boundary Fusion of a Two-Base Lens-Space Beta Integral

Hainan Zhao
Independent Researcher
hainzhao@gmail.com

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Abstract

We reformulate the remaining dimension-six case of the formal Twisted Convolution Conjecture of Appleby–Flammia–Kopp as one boundary theorem for a two-base general modular quantum-dilogarithm packet. The dimension-six principal-ghost quotient is identified with the two-gamma kernel in the degenerate Sarkissian–Spiridonov lens-space beta transform. In the upper half-plane its natural bases are $e^{2\pi i\tau}$ and $e^{2\pi i A_6\tau}$ and must not be fused. At the real-multiplication fixed point $\beta_6 + \beta_6^{-1} = 5$ they coalesce arithmetically and give the previously isolated well-poised ${}_2\psi_2$ packet at argument $-q$. The identical calculation in the proved dimension-five case lands instead on the summable $+q$ locus, while the independent dimension-four even-wrap control returns to $-q$. We verify the interior identity as a meromorphic spectral identity, specialize the published beta evaluation exactly, and show that the undeformed endpoint contour is not absolutely convergent because its quadratic decay cancels. We define the finite part by an interior tilted contour and prove tilt-independence by Cauchy's theorem in the pole-free strip. An exact all-characteristic calculation gives $A_6 \equiv I \pmod 6$, $\psi^2(A_6) = -1$, and equality of the Kopp and AFK multipliers in all thirty-six cases. Consequently one explicitly stated analytic fusion-continuity conjecture implies both formal dimension-six shifts. We do not claim that conjecture here.

1 Statement and proof status

Put

$$\beta = \frac{5+\sqrt{21}}{2}, \qquad A = \begin{pmatrix} 115 & -24 \\ 24 & -5 \end{pmatrix}. \tag{1}$$

Then

$$\beta^2 - 5\beta + 1 = 0, \qquad A\beta = \beta, \qquad \beta + \beta^{-1} = 5. \tag{2}$$

The first equality is the arithmetic trace relation; throughout this paper it is part of the boundary condition, not a generic removable limit.

The finite part of the dimension-six calculation is already exact. There are two formal shifts, the reconstructed matrix has trace one, and all 225 rank-two minors vanish after imposing the certified algebraic packet. What is missing is the analytic identification of that packet with the convention-matched AFK cocycle values. The purpose of this note is to make the remaining analytic statement independent of the finite certificates.

Parametrize the attracting half of the A-axis by

$$\gamma(s) = \frac{5}{2} + \frac{\sqrt{21}}{2}\frac{1-s^2}{1+s^2} + i\frac{\sqrt{21}s}{1+s^2}, \qquad s > 0.$$

Then $\gamma(s) \to \beta$ as $s \downarrow 0$. Kopp's Definition 4.7 chooses the positive stabilizer generator by this attracting direction and observes that it translates the same geodesic toward β [2]. Definition 4.9 defines the RM stable value by evaluation of the meromorphic continuation at the fixed point, and Proposition 7.20 realizes it as a limit from the upper half-plane.

Let $\mathcal{P}_j(\tau)$, $j = 0, 1, 2$, denote the primitive ray-class logarithms of the spectral periodization in the norm-37 Frobenius order, with logarithm branches continued from the two-base chamber along γ, and put

$$\mathcal{R}_1(\tau) = \mathcal{P}_0(\tau) + \zeta_6 \mathcal{P}_1(\tau) + \zeta_6^2 \mathcal{P}_2(\tau).$$

For τ in the interior convergence chamber, put $S_\tau = \{y : 0 < \Re y < \Re Q_\tau\}$. If

$$C_h = \{c + h(t) + it : t \in \mathbb{R}\} \subset S_\tau$$

is a C^1 graph homotopic to $c + i\mathbb{R}$ through graphs on which the two-base kernel has one common integrable majorant, define its *tilted value* by integration over C_h, followed by the exact helical periodization.

Proposition 1 (Tilt independence)**.** *The interior tilted value is independent of the admissible graph C_h.*

Proof. Truncate two admissible graphs at imaginary heights $\pm T$ and join their endpoints inside S_τ. The kernel is holomorphic in the region between them because its pole cones lie on opposite sides of the strip. Cauchy's theorem makes the integral around the resulting boundary zero. The common interior exponential majorant makes both joining integrals tend to zero as $T \to \infty$. The two graph integrals are equal. Absolute local uniform convergence permits the same argument after helical periodization. □

Thus $\mathcal{P}_j(\tau)$ may equivalently be defined by any admissible tilt, and

$$\mathrm{FP}_\beta\, \mathcal{P}_j := \lim_{s \downarrow 0} \mathcal{P}_j(\gamma(s)),$$

if it exists, has no remaining contour-choice ambiguity.

For each j, let U_j^{fus} be the boundary value obtained by first taking the meromorphic spectral periodization with its certified level-24 label, then imposing the fixed-point relation $\widetilde{q}_M = q_M$, and evaluating the resulting expression through the 24-factor continuation rather than the collapsed bilateral series. The logarithm is the continuation of the branch used for $\mathcal{P}_j$. Define

$$\mathrm{Fus}_5\, \mathcal{R}_1 := \log U_0^{\mathrm{fus}} + \zeta_6 \log U_1^{\mathrm{fus}} + \zeta_6^2 \log U_2^{\mathrm{fus}}.$$

The subscript 5 records the trace equation $\beta + \beta^{-1} = 5$, not a dimension.

Conjecture 2 (Minimal fusion-continuity, MFC$_6$)**.** *The scalar $\mathcal{R}_1(\gamma(s))$ has a finite limit as $s \downarrow 0$, and meromorphic spectral periodization commutes in this primitive component with trace-five base fusion, preserving the norm-37 lens/Frobenius label:*

$$\lim_{s \downarrow 0} \mathcal{R}_1(\gamma(s)) = \mathrm{Fus}_5\, \mathcal{R}_1.$$

No uniform convergence, Hölder estimate, or two-sided continuity is assumed. Since A translates the same oriented geodesic toward its attracting fixed point, existence of the limit makes it A-invariant; invariance is not an additional hypothesis. The exceptional zero frequency is also not part of MFC$_6$: it is supplied separately by the exact AFK endpoint $-4\sqrt{7}$. Its tilted Fresnel/Abel value gives the reciprocal pair $-2\sqrt{7} \pm 3\sqrt{3}$, whose trace is $-4\sqrt{7}$.

Theorem 3 (Analytic-to-Stark bridge)**.** *Assume Conjecture 2. Let* $G = \mathrm{Cl}_{(6)\infty_2}(\mathbb{Q}(\sqrt{21})) = \langle g\rangle \simeq C_6$*, let* $\chi_1(g) = \zeta_6$*, and let* r_j *be the logarithms of the certified algebraic ray-unit orbit in the arithmetic Frobenius order. Then*

$$L'_S(0,\chi_1) = r_0 + \zeta_6 r_1 + \zeta_6^2 r_2. \tag{3}$$

Thus a flow-invariant analytic regularity statement at the A_6 *fixed point implies the convention-fixed rank-one Stark identity over* $\mathbb{Q}(\sqrt{21})$*.*

Theorem 4 (Conditional dimension-six closure)**.** *Assume Conjecture 2. Then* 0 *and* 1 *are shifts for every admissible dimension-six tuple in the sense of AFK [1, Definition 1.34 and equation (1.49)].*

The implication in Theorem 4 is proved in Section 6. Conjecture 2 remains open. In particular, the theorem is not an unconditional proof of the dimension-six TCC.

2 The honest two-base packet

For the general modular quantum dilogarithm of Sarkissian–Spiridonov [3], use

$$(p,k,r,s) = (-115, 24, 5, 24), \qquad pr + ks = 1. \tag{4}$$

Their equations (5)–(8) give the two bases

$$q_M = e^{2\pi i\tau}, \qquad \widetilde{q}_M = e^{2\pi i A\tau}. \tag{5}$$

They are distinct in the upper half-plane. Their equation (15) factors the same general modular function into 24 standard Faddeev factors. If $\rho = 24\tau - 5$, the bases in each standard factor are

$$q_S = e^{2\pi i\rho}, \qquad \widetilde{q}_S = e^{-2\pi i/\rho}. \tag{6}$$

Equations (4) and (5) are different modular pairs away from the boundary. Conflating them gives the false equal-base interior model.

Proposition 5 (Interior factorization)**.** *In the chamber in which both product bases have modulus less than one, the direct general modular gamma in (4) equals its* 24*-factor Faddeev continuation. Periodizing the degenerate two-gamma kernel over the helical quotient*

$$\big(\mathbb{R}\times\mathbb{Z}/24\mathbb{Z}\big)/\langle(\omega_1-\omega_2, 6)\rangle \tag{7}$$

gives three absolutely and locally uniformly convergent bibasic bilateral classes for each level-six frequency. The resulting identity is valid as a meromorphic spectral identity.

Proof. The factorization is equation (15) of [3] specialized by (3). For a frequency $(a,b) \bmod 6$ and alias $z\in\mathbb{Z}$, direct reduction gives

$$N_z = a + 2 - 6z, \qquad \ell_z = b - 6z, \tag{8}$$

and

$$\frac{\alpha_z}{4\tau-1} = \frac{4b-5a}{3} + 2z. \tag{9}$$

Thus $-5(N_z-2) \equiv a$ and $\ell_z \equiv b \pmod 6$. Three alias steps are one helical translation at level 24. The functional-equation ratio is bibasic and has geometric tails in the stated chamber. Finite heads and the bound

$$\frac{|a|^{M+1}}{(M+1)(1-|a|)(1-|q|)} \tag{10}$$

for the omitted logarithmic q-Pochhammer tail prove local uniform convergence. This justifies the spectral reindexing. The exact integer ledger and Arb enclosures are reproduced by `scripts/dimension_six_two_base_lens.py` and `scripts/dimension_six_interior_factorization_audit.py`. □

At $\tau = \beta$, both pairs fuse, but for different reasons: $A\beta = \beta$ for (4), while $\beta + \beta^{-1} = 5$ makes the bases in (5) equal. Exact substitution in the term ratio gives

$$-q \, \frac{(1-x)(1+w^{-1}x)}{(1+qx)(1-qw^{-1}x)}. \tag{11}$$

This is the well-poised ${}_2\psi_2$ packet at argument $-q$ found in the finite Zak descent. Equation (11) is a boundary identity only.

3 The dimension-five control

For dimension five,

$$\beta_5 = 2+\sqrt{3}, \qquad A_5 = \begin{pmatrix} 56 & -15 \\ 15 & -4 \end{pmatrix}, \qquad \beta_5 + \beta_5^{-1} = 4. \tag{12}$$

Running the same two-base construction gives at fusion

$$q \, \frac{(1-x)(1-w^{-1}x)}{(1-qx)(1-qw^{-1}x)}. \tag{13}$$

Proposition 6 (Summability dichotomy). *The proved dimension-five packet lies on the positive-argument closed ${}_2\psi_2$ locus (13). The dimension-six packet lies on its sign-reflected neighbor (11).*

Proof. The sign follows from the exact even-wrap and lens-label reductions. For dimension five the direct two-base function and its factorization are enclosed at three interior points, all six bibasic classes converge with rigorous tails, and the boundary double-sine square encloses the independently proved algebraic root

$$3.89086171394307925533764395 96\ldots.$$

The executable proof is `scripts/dimension_five_two_base_calibration.py`. The dimension-four control independently fixes lens level 8 and phase level 16. Exact exponent reduction also gives

$$-q \, \frac{(1-x)(1-ix)}{(1+qx)(1+iqx)},$$

so dimension four, like dimension six, lies at argument $-q$; see `scripts/dimension_four_two_base_calibration.py`. The exceptional dimension-six zero mode is independently normalized by

$$\nu_{\text{aux}} + \nu_{\text{aux}}^{-1} = -4\sqrt{7}.$$

Thus the complete calibration triangle is

$$d = 4 : -q, \qquad d = 5 : +q, \qquad d = 6 \text{ zero mode} : -4\sqrt{7}.$$

□

This control is decisive: the minus sign in (11) is not a dispensable normalization, and replacing it by the summable plus sign would erase the dimension-six obstruction.

4 Exact beta evaluation and its residue

Sarkissian–Spiridonov's unnumbered beta-evaluation theorem immediately preceding their equation (58), followed by their degeneration (66), applies meromorphically after the specialization

$$\omega_1 = 24\beta - 5 = \beta^3, \quad \omega_2 = 1, \quad Q = \omega_1 + \omega_2, \quad g = Q, \quad \ell = 0. \tag{14}$$

The continuous phase coefficient is

$$p - k(1-s) = -115 - 24(1-24) = 437. \tag{15}$$

For the Fourier variables (α, N), equation (66) becomes

$$\begin{aligned} \sum_{m \bmod 24} \int e^{\pi i m\, 437(2N-4)/24} e^{\pi i \alpha(2y-Q)/(24\omega_1)} \Gamma_M(y,m)\Gamma_M(Q-y,-m) \frac{dy}{i\sqrt{\omega_1}} \\ = 24\Gamma_M(-\alpha, 4-N)\Gamma_M(\alpha,N)\Gamma_M(Q,0), \end{aligned} \tag{16}$$

as a meromorphic Fourier identity.

Proposition 7 (Conservation of orientation)**.** *The right side of (15) does not reduce, by the standard double-sine reflection and period-shift relations, to an unoriented norm identity. The irreducible oriented residue is*

$$\Gamma_M(-\alpha, 4-N)\Gamma_M(\alpha, N). \tag{17}$$

Proof. The reflection partner of $\Gamma_M(\alpha, N)$ is $\Gamma_M(Q-\alpha, 4-N)$, not $\Gamma_M(-\alpha, 4-N)$. Shifting by Q changes the discrete label by $r - 1 = 4$, which is not zero modulo 24. Canonical reduction therefore leaves (16). The full parameter and relation-basis audit is `scripts/dimension_six_ss_evaluation_audit.py`. □

Thus (15) is a genuine integral-transform identity but not, on its own, the new finite multiplicative relation required to identify the dimension-six algebraic packet.

5 Why the endpoint contour does not close the proof

Proposition 8 (Endpoint obstruction)**.** *At (13), the undeformed vertical contour in (15) is not absolutely convergent for any real Fourier frequency α.*

Proof. For $y = i\lambda$, the two asymptotic formulas [3, equations (40)–(41)] give

$$\Gamma_M(y,m) \sim Z(m)e^{-\pi i B_{2,2}(y)/48}, \quad \Gamma_M(Q-y,-m) \sim Z(-m)^{-1}e^{+\pi i B_{2,2}(Q-y)/48}. \tag{18}$$

Since $B_{2,2}(Q-y) = B_{2,2}(y)$, the quadratic decay cancels and the kernel tends to the nonzero constant $Z(m)/Z(-m)$. The remaining phase has modulus

$$\exp\left(-\frac{2\pi\alpha\lambda}{24\omega_1}\right). \tag{19}$$

It grows at one end when $\alpha \neq 0$ and fails to decay at either end when $\alpha = 0$. □

There is nevertheless no finite pole pinch at $g = Q$. The poles of the first gamma factor lie in the cone at and to the left of 0, while those of the reflected factor lie in the cone at and to the right of g. At $g = Q > 0$ the cones remain separated by Q. Hence no finite-pinch residue correction occurs; the obstruction is entirely at imaginary infinity.

Likewise, the fused bilateral series has $cd/(ab) = q^2$ and $z = -q$. Slater's strict convergence annulus $|q|^2 < |q| < 1$ collapses at the boundary. Neither the original contour nor the boundary bilateral sum proves Conjecture 2; equation (15) survives only as a meromorphic continuation.

Component type and the exact small divisor

For finite frequency (a, b) mod 6, choose the centered lift

$$s_{a,b} \equiv 4b - 5a \pmod 6, \qquad s_{a,b} \in \{-3, -2, -1, 0, 1, 2\}.$$

Exactly six components have $s_{a,b} = 0$; they are purely oscillatory and admit the Fresnel/Abel prescription. The remaining thirty grow at one end of the vertical contour and require the interior strip and its tilted finite part. Reversing the convention for residue three changes six growth orientations but not this $6 + 30$ split.

Put $\Delta(\tau) = \tau + \tau^{-1} - 5$. Direct calculation gives

$$A\tau - \tau = -\frac{24\tau}{24\tau - 5}\Delta(\tau). \tag{19a}$$

Equivalently,

$$\gamma(s) = \frac{\beta + \beta^{-1}s^2 + i\sqrt{21}s}{1 + s^2},$$

and therefore

$$\Delta(\gamma(s)) = i\frac{21}{\beta}s + O(s^2), \qquad 2\pi|\Im(A\gamma(s) - \gamma(s))| = 2\pi\sqrt{21}(1 - \beta^{-6})s + O(s^2). \tag{19b}$$

This is the degenerating two-base decay coefficient.

The arithmetic at the same point is explicit:

$$\beta = [4; \overline{1, 3}], \qquad (n\beta - m)(n\beta^{-1} - m) = m^2 - 5mn + n^2.$$

Taking m nearest to $n\beta$ and using the nonzero integral norm gives

$$\|n\beta\| \geq \frac{1}{\sqrt{21}\,n + \frac{1}{2}}, \qquad |1 - e^{2\pi i n\beta}| \geq \frac{4}{\sqrt{21}\,n + \frac{1}{2}}. \tag{19c}$$

Thus exponential damping of rate $\kappa(s) \asymp s$ competes with quadratic-irrational divisors of order $1/n$ in the transition range $n \asymp 1/s$. This is the sharp small-divisor form of MFC_6.

Proposition 9 (Analytic stratum audit)**.** *For both formal shifts, the six Fresnel frequency modes coincide exactly with the six q-gamma singular-cancellation modes. They do not coincide with the conductor-lowered arithmetic stratum.*

Proof. The Fresnel condition is $4b - 5a \equiv 0 \pmod 6$. Inverting the two exact TCC frequency maps sends its six solutions respectively to

$$\{(0, 0), (4, 1), (2, 2), (0, 3), (4, 4), (2, 5)\}$$

and

$$\{(0, 0), (5, 5), (4, 4), (3, 3), (2, 2), (1, 1)\}.$$

These are exactly the respective q-gamma cancellation patterns. Each set has denominator counts

$$(1, 1, 2, 2) \quad \text{at denominators } (1, 2, 3, 6).$$

The thirty growing modes have counts $(0, 2, 6, 22)$. Only one point of the proved three-point modulus-three orbit is Fresnel for either shift. The executable exact comparison is `scripts/dimension_six_fresnel_stratum_audit.py`. $\square$

A standalone sufficient estimate

For $\tau \in \mathbb{H}$, put

$$\omega = 24\tau - 5, \quad q = e^{2\pi i\tau}, \quad \widetilde{q} = e^{2\pi i A\tau},$$

and define

$$G_\tau(\mu, m) = \frac{(\widetilde{X}(\mu, m); \widetilde{q})_\infty}{(X(\mu, m); q)_\infty},$$

where

$$X(\mu, m) = e^{2\pi i(\mu+m)/24}, \quad \widetilde{X}(\mu, m) = \widetilde{q}\, e^{2\pi i(\mu+115m\omega)/(24\omega)}.$$

With $D = 4\tau - 1$, let

$$\alpha_z = D(4b - 5a)/3 + 2Dz, \qquad N_z = a + 2 - 6z,$$

$$K_{a,b}(z; \tau) = G_\tau(\alpha_z, N_z) G_\tau(-\alpha_z, 4 - N_z),$$

$$\mathcal{S}_{a,b,r}(s) = \sum_{k\in\mathbb{Z}} K_{a,b}(r + 3k; \gamma(s)), \qquad r = 0, 1, 2.$$

One alias period has multiplier $-q$; at the boundary the oscillations are therefore $(-1)^k e^{2\pi i k\beta}$ and concentrate at $|k| \asymp s^{-1}$.

Conjecture 10 (Quantitative boundary fusion, $\mathrm{BF}_6(\eta)$)**.** *For some $\eta, C, s_0 > 0$, uniformly over the thirty pairs $4b - 5a \not\equiv 0 \pmod 6$ and $r = 0, 1, 2$,*

$$\left|\mathcal{S}_{a,b,r}(s) - \mathcal{S}^{\mathrm{fus}}_{a,b,r}\right| \le C\,|\Delta(\gamma(s))|^\eta, \qquad 0 < s < s_0, \tag{19d}$$

where the fused value is the inherited 24-factor Abel–Fresnel limit.

A Dini modulus tending to zero would suffice instead of a power. Conjecture 10 implies MFC_6 and hence Theorems 3 and 4. The completely standalone formulation is `docs/dimension-six-standalone-estimate.md`.

Conditioning slopes

After dividing the relative Arb radius by the requested q-Pochhammer tolerance, the pinned asymptotic windows give

d	$\mathrm{Tr}(\beta)$	$\sqrt{\operatorname{disc} K}$	$\dfrac{d\log_{10} C_d}{d(1/s)}$	$\dfrac{d\log_{10} C_d}{d(1/\lvert\tau-\beta\rvert)}$
4	3	$\sqrt{5}$	0.643602	1.43948
6	5	$\sqrt{21}$	2.803972	12.8721

(19e)

with $R^2 = 0.999933$ and 0.999996, respectively, and replication at a second precision. This is essential-exponential conditioning in $1/s$ for the factorized-continuation implementation. It is not an intrinsic exponent of BF_6. The dimension-four value is proved despite the same pathology; its slope is about 4.36 times smaller. See `scripts/dimension_six_conditioning_comparison.py`.

6 Stabilizer ledger and conditional closure

Lemma 11 (All multipliers)**.** *The matrix A fixes every characteristic modulo 6, satisfies $\psi^2(A) = -1$, and for every $\boldsymbol{r} = (a/6, b/6)$,*

$$(\psi^{-2}\chi_{\boldsymbol{r}}^{-1})(A) = \Phi^2_{a,b}. \tag{20}$$

Proof. The first assertion is $A \equiv I \pmod 6$. The Rademacher invariant is 6, so

$$\psi^2(A) = e^{\pi i 6/6} = -1.$$

Substitution of $\boldsymbol{r}$ in Kopp's theta-character formula and reduction modulo one agrees with the square of

$$\Phi_{a,b} = (-1)^{6+7(1+a)(1+b)} e^{-\pi i 6/12} \tau_6^{-(a^2-5ab+b^2)} \tag{21}$$

in all 36 cases. This is exact rational arithmetic; see `scripts/dimension_six_stabilizer_ledger.py`. □

Proof of Theorem 4. Conjecture 2 identifies the two-base boundary packet with the AFK/Kopp packet characteristic by characteristic. Lemma 11 shows that this identification has the required stabilizer phase. The exact finite frequency maps then turn the two formal TCC sums into the two reconstructed matrices. Their trace is one and all 225 rank-two minors vanish, so each has rank one and is idempotent. The matrix-to-TCC Fourier bridge gives both shifts for the canonical tuple. Integral covariance under $\mathrm{GL}_2(\mathbb{Z})$, applied also to the inverse transformation, gives equality of shift sets and transports the result to every admissible dimension-six form. □

Proof of Theorem 3. The exact characteristic-to-ray bridge identifies the fused boundary packet with the differenced derivatives $D_j = Z'_{(6)\infty_2}(0, g^j)$ in the certified Frobenius order. Projecting to the primitive order-six Fourier line gives

$$L'_S(0, \chi_1) = r_0 + \zeta_6 r_1 + \zeta_6^2 r_2. \tag{22}$$

No TCC identity or minor vanishing is used in this projection. □

Grade-2 equivalence and Grade-3 attack surface

The rigid endpoint content of Conjecture 2 is reduction-equivalent to (22). Write

$$\Lambda_1 = A_0 + \zeta_6 B_0, \qquad Q_3 = D_0 - D_1 + D_2.$$

The reverse reduction is

$$D_0 = \frac{Q_3 + 2A_0 + B_0}{3}, \qquad D_1 = \frac{-Q_3 + A_0 + 2B_0}{3}, \qquad D_2 = \frac{Q_3 - A_0 + B_0}{3}. \tag{23}$$

Indeed, (22) fixes Λ_1, conjugation fixes Λ_5, and the proved conductor-three identity fixes Q_3. Equation (23), sign-class reciprocity, shift/reflection/duplication, conductor lowering, and Lemma 11 recover all thirty-six endpoint values. Conversely, primitive Fourier projection gives (22). Neither direction uses TCC or a minor certificate.

The literal MFC_6 assertion is not derived from (22) by this standard basis: equation (22) does not establish existence of the geodesic limit or commutation of periodization with fusion. Thus the endpoint value is Grade-2 equivalent, but the converse for the full regularity hypothesis is open. Its family formulation has a genuinely richer attack surface: one may differentiate in τ, deform the contour, track pinches and residue jumps, vary the lens label, iterate the A_6 return map, and use badly-approximable-point estimates. This analytic structure is absent from the rigid equality (22).

7 Scope and reproducibility

The verified content of this note is the two-base reformulation, the dimension-five/dimension-six dichotomy, the meromorphic Sarkissian–Spiridonov specialization, the endpoint exclusion, and the finite multiplier ledger. The unconditional TCC is already proved in dimensions four, five, seven, and eight in the companion papers [4, 5]. Dimension six remains conditional on Conjecture 2.

The cycle-by-cycle state and all negative-result gates are recorded in `docs/dimension-six-state-notes-v3` `md`. The executable certificate packet is regenerated by `scripts/generate_dimension_six_amendment_certificates.sh`. The zero-frequency tilted/Fresnel normalization gives the reciprocal pair $-2\sqrt{7} \pm 3\sqrt{3}$ and hence the independently enclosed trace $-4\sqrt{7}$. This is a normalization calibration only; it does not imply the nonzero oriented fusion-continuity assertion.

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